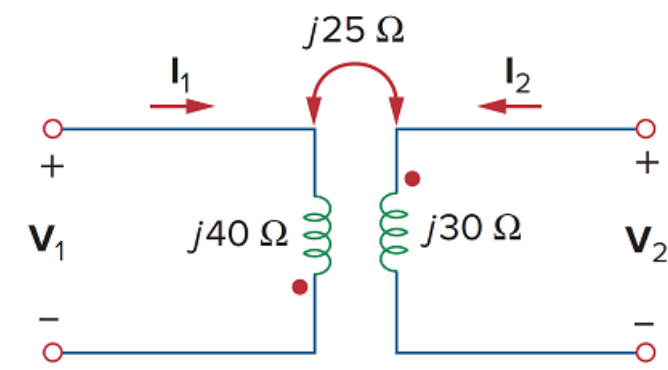


Problem

Determine an equivalent T-section that can be used to replace the transformer in Fig. 13.88.

Figure 13.88



Step-by-step solution

Step 1 of 2

Let $\omega = 1$.

$$L_a = L_1 - (-M) = 40 + 25 = 65 \text{ H}$$

$$L_b = L_2 + M = 30 + 25 = 55$$

$$L_c = -M = -25$$

Thus the T – section is as shown below.

Step 2 of 2

